



l : length pivot to com
 L : total length
 I : MOI wrt com
 V : velocity of link com
 C : coeff of damping

Link 1

$$x_1 = x_c - l_1 \sin \theta_1$$

$$y_1 = l_1 \cos \theta_1$$

$$\dot{x}_1 = \frac{dx_1}{dt} = \frac{\partial x_1}{\partial x_c} \frac{dx_c}{dt} + \frac{\partial x_1}{\partial \theta_1} \frac{d\theta_1}{dt}$$

$$= \dot{x}_c - l_1 \cos \theta_1 \dot{\theta}_1$$

$$\dot{y}_1 = \frac{dy_1}{dt} = \frac{\partial y_1}{\partial \theta_1} \frac{d\theta_1}{dt}$$

$$= -l_1 \sin \theta_1 \dot{\theta}_1$$

$$V_1^2 = \dot{x}_1^2 + \dot{y}_1^2$$

$$= (\dot{x}_c - l_1 \cos \theta_1 \dot{\theta}_1)^2 + (-l_1 \sin \theta_1 \dot{\theta}_1)^2$$

$$= \dot{x}_c^2 - 2\dot{x}_c l_1 \cos \theta_1 \dot{\theta}_1 + l_1^2 \cos^2 \theta_1 \dot{\theta}_1^2 + l_1^2 \sin^2 \theta_1 \dot{\theta}_1^2$$

$$= \dot{x}_c^2 - 2\dot{x}_c l_1 \cos \theta_1 \dot{\theta}_1 + l_1^2 \dot{\theta}_1^2$$

Link 2

$$x_2 = x_c - l_1 \sin \theta_1 - l_2 \sin \theta_2$$

$$y_2 = l_1 \cos \theta_1 + l_2 \cos \theta_2$$

$$\dot{x}_2 = \frac{\partial x_2}{\partial x_c} \dot{x}_c + \frac{\partial x_2}{\partial \theta_1} \dot{\theta}_1 + \frac{\partial x_2}{\partial \theta_2} \dot{\theta}_2$$

$$= \dot{x}_c - l_1 \cos \theta_1 \dot{\theta}_1 - l_2 \cos \theta_2 \dot{\theta}_2$$

$$\dot{y}_2 = \frac{\partial y_2}{\partial \theta_1} \dot{\theta}_1 + \frac{\partial y_2}{\partial \theta_2} \dot{\theta}_2$$

$$= -l_1 \sin \theta_1 \dot{\theta}_1 - l_2 \sin \theta_2 \dot{\theta}_2$$

$$V_2^2 = (\dot{x}_c - l_1 \cos \theta_1 \dot{\theta}_1 - l_2 \cos \theta_2 \dot{\theta}_2)^2 + (-l_1 \sin \theta_1 \dot{\theta}_1 - l_2 \sin \theta_2 \dot{\theta}_2)^2$$

$$= \dot{x}_c^2 - 2\dot{x}_c l_1 \cos \theta_1 \dot{\theta}_1 - 2\dot{x}_c l_2 \cos \theta_2 \dot{\theta}_2 - 2l_1 l_2 \cos \theta_1 \cos \theta_2 \dot{\theta}_1 \dot{\theta}_2$$

$$+ l_1^2 \cos^2 \theta_1 \dot{\theta}_1^2 + l_1^2 \sin^2 \theta_1 \dot{\theta}_1^2 + l_2^2 \cos^2 \theta_2 \dot{\theta}_2^2 + l_2^2 \sin^2 \theta_2 \dot{\theta}_2^2$$

$$+ 2l_1 l_2 \sin \theta_1 \sin \theta_2 \dot{\theta}_1 \dot{\theta}_2$$

$$= \dot{x}_c^2 - 2L_1 \dot{x}_c \dot{\theta}_1 \cos \theta_1 - 2l_2 \dot{x}_c \dot{\theta}_2 \cos \theta_2 + l_1^2 \dot{\theta}_1^2 + 2L_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos \theta_1 \cos \theta_2 + 2L_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin \theta_1 \sin \theta_2 + l_2^2 \dot{\theta}_2^2$$

$$= \dot{x}_c^2 - 2L_1 \dot{x}_c \dot{\theta}_1 \cos \theta_1 - 2l_2 \dot{x}_c \dot{\theta}_2 \cos \theta_2 + l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2L_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2)$$

$$= \dot{x}_c^2 - 2L_1 \dot{x}_c \dot{\theta}_1 \cos \theta_1 - 2l_2 \dot{x}_c \dot{\theta}_2 \cos \theta_2 + l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2L_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)$$

Potential:

$$U = g[m_1 y_1 + m_2 y_2]$$

$$= g[m_1 l_1 \cos \theta_1 + m_2 (l_1 \cos \theta_1 + l_2 \cos \theta_2)]$$

Kinetic:

$$T = T_c + T_1 + T_2$$

$$T_c = \frac{1}{2} M \dot{x}_c^2$$

$$T_1 = \frac{1}{2} m_1 V_1^2 + \frac{1}{2} I_1 \dot{\theta}_1^2$$

$$T_2 = \frac{1}{2} m_2 V_2^2 + \frac{1}{2} I_2 \dot{\theta}_2^2$$

$$\mathcal{L} = T - U$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_j$$

$$\text{OE1: } \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_c} - \frac{\partial \mathcal{L}}{\partial x_c} = F - C_c \dot{x}_c$$

$$\frac{\partial \mathcal{L}}{\partial x_c} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_c} = f_1(\dot{x}_c, x_c, \dot{\theta}_1, \theta_1, \dot{\theta}_2, \theta_2)$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_c} = \frac{\partial f_1}{\partial \dot{x}_c} \dot{x}_c + \frac{\partial f_1}{\partial \dot{\theta}_1} \dot{\theta}_1 + \frac{\partial f_1}{\partial \dot{\theta}_2} \dot{\theta}_2$$

$$\text{eg } \frac{d}{dt} (m_1 l_1 \cos \theta_1 \dot{x}_c) = m_1 l_1 \left[\cos \theta_1 \frac{d\dot{x}_c}{dt} + \dot{x}_c \frac{d \cos \theta_1}{dt} \right]$$

$$= m_1 l_1 \left[\cos \theta_1 \frac{\partial \dot{x}_c}{\partial \dot{x}_c} \frac{d\dot{x}_c}{dt} + \dot{x}_c \frac{\partial \cos \theta_1}{\partial \theta_1} \frac{d\theta_1}{dt} \right]$$

$$= \cos \theta_1 \cdot 1 \cdot \ddot{x}_c + \dot{x}_c \cdot (-\sin \theta_1) \cdot \dot{\theta}_1$$

$$= \nabla_{\dot{x}_c} \cdot \ddot{x}_c + \nabla_{\theta_1} \cdot \dot{\theta}_1$$

$$\text{OE2: } \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} - \frac{\partial \mathcal{L}}{\partial \theta_1} = -C_1 \dot{\theta}_1 + C_2 (\dot{\theta}_2 - \dot{\theta}_1)$$

$$\text{OE3: } \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} - \frac{\partial \mathcal{L}}{\partial \theta_2} = -C_2 (\dot{\theta}_2 - \dot{\theta}_1)$$